

Independence, Basis and Dimension

Which funds are genuinely different? What is the smallest menu that still reaches everything? And how many independent directions does a menu really have? Three words answer it: independence, basis, dimension.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. Conditions and important concepts

13 short definitions: independence, dependence, span, basis, dimension, bases of $C(A)$ and $N(A)$.

2. Procedure

How to test independence, and how to find a basis and the dimension.

3. 2×2 examples

Nifty and bonds (independent); Nifty and its twin (dependent).

4. 3×3 examples

Nifty, bonds, midcap (a basis of $\mathbb{R}^3$); Nifty, bonds, balanced (dependent: bases of $C(A)$ and $N(A)$).

5. Bigger example: five funds

Five funds, three genuinely different directions. With a fund picker.

6. Connection: rank, dimension and $n - r$

Every fund is either a new direction or a redundancy.

7. Final memory table

Every concept, its test and an example.

METHODS IN THIS LESSON

Independence

Independence test

Span

Basis

Dimension

Basis of $C(A)$

Basis of $N(A)$

PART 2 Prerequisites

Click any link to review it, then come back.

- Note 1: $Ax = 0$ (Lesson 7): R, pivots, rank, free variables, special solutions.

- The column space (Lesson 5) and the null space (Lesson 6).
- Arrow-Debreu building blocks (Lesson 3).

PART 3 Explanation

THE FUNDS USED IN THE EXAMPLES

Fund	Column: what ₹1 lakh does to (budget, return, exposure)
Nifty	(1, 12, 1)
Nifty ETF (a twin of Nifty)	(1, 12, 1)
Bonds	(1, 7, 0)
Balanced (half Nifty, half bonds)	(1, 9.5, 0.5)
Midcap	(1, 15, 1.2)

In this lesson each fund is a **vector** (its column). The questions are about the funds themselves: are they genuinely different, and how many do you really need?

1. Conditions and important concepts

1. What is linear independence?

DEFINITION · LINEARLY INDEPENDENT VECTORS

Vectors $v_1, v_2, \dots, v_n$ are **independent** if $c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$ happens **only** when every c is 0.

In money terms:

Independent funds: no fund can be rebuilt from the others.

The only mix of them that changes nothing is "hold none of them".

2. What is dependence?

Vectors are **dependent** if some mix with not-all-zero c 's gives 0. Example: Nifty, bonds, balanced:

$$-\frac{1}{2} \begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 9.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -0.5 - 0.5 + 1 \\ -6 - 3.5 + 9.5 \\ -0.5 + 0 + 0.5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

That mix is a do-nothing trade, and it shows $\text{balanced} = \frac{1}{2} \text{Nifty} + \frac{1}{2} \text{bonds}$. So:

Dependent = a do-nothing trade exists = some fund is a mix of the others.

3. The independence test

Put the vectors as the columns of a matrix A . A mix $c_1 v_1 + \dots + c_n v_n$ is just Ac , so the question "is there a non-zero c with $Ac = 0$?" is the null space question from Note 1:

RULE · THE INDEPENDENCE TEST

The columns of A are independent $\iff N(A) = \{0\} \iff$ no free variables $\iff$ every column is a pivot column $\iff \text{rank} = n$.

4. The picture

Two vectors are dependent exactly when they lie on the same line (one is a multiple of the other): Nifty and its twin. **Three** vectors in $\mathbb{R}^3$ are dependent exactly when they lie in the same plane: Nifty, bonds and balanced all lie in the plane $R = 7B + 5E$ (Lesson 5).

5. Too many vectors are always dependent

More than m vectors in $\mathbb{R}^m$ are always dependent: $n > m$ means a free variable (Note 1). Example: any 4 funds measured by 3 rules.

$$n > m \Rightarrow \text{dependent}$$

6. What is span?

DEFINITION · SPAN

Vectors $v_1, \dots, v_k$ **span** a space when the space consists of **all** their combinations. The columns of A span the column space $C(A)$.

In money terms: the span is every target the funds can reach. Nifty and bonds span the plane $R = 7B + 5E$.

7. What is a basis?

DEFINITION · BASIS

A **basis** for a space is a set of vectors that are (1) **independent** and (2) **span** the space.

Basis = reaches everything (span) + no waste (independent).

In money terms: the smallest complete menu. Drop any fund and you lose something; add any fund and it's redundant.

8. The standard basis

The simplest basis of $\mathbb{R}^3$:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Independent: $c_1(1, 0, 0) + c_2(0, 1, 0) + c_3(0, 0, 1) = (c_1, c_2, c_3)$, which is 0 only if every c is 0.

Spanning: any (b_1, b_2, b_3) is b_1 of the first + b_2 of the second + b_3 of the third. In markets these are the **Arrow-Debreu securities** from Lesson 3: each pays 1 in exactly one scenario.

9. A space has many bases

Nifty, bonds, midcap are also a basis of $\mathbb{R}^3$ (3 pivots, worked below). In general:

n vectors in $\mathbb{R}^n$ form a basis $\iff$ their matrix is invertible $\iff n$ pivots

10. Dimension

DEFINITION · DIMENSION

Every basis of a space has the **same number** of vectors. That number is the **dimension** of the space.

$\mathbb{R}^3$ has dimension 3 (both bases above have 3 vectors). The plane $R = 7B + 5E$ has dimension 2. In money terms: the number of genuinely different directions you can trade.

11. Basis of the column space $C(A)$

METHOD · BASIS OF $C(A)$ = THE PIVOT COLUMNS OF A

Reduce A to R , see which columns have pivots, then take **those columns of the original A** . They form a basis of $C(A)$, and $\dim C(A) = \text{rank}$.

IMPORTANT

Use the columns of A , **not** the columns of R . Row operations keep the null space but change the column space (Note 1). An example is in the 3×3 section.

12. Basis of the null space $N(A)$

METHOD · BASIS OF $N(A)$ = THE SPECIAL SOLUTIONS

The special solutions (one free variable = 1, the others 0) are independent and span $N(A)$. So they are a basis, and $\dim N(A) = n - \text{rank}$.

Why independent: each special solution has a 1 in its own free slot where all the others have 0, so no mix of them can be zero unless every coefficient is.

13. Rank vs dimension

Matrices have a **rank**; spaces have a **dimension**. They meet here:

$$\text{rank}(A) = \text{number of pivot columns} = \dim C(A)$$

2. Procedure

A. Are these vectors independent?

Step	What to do
1	Put the vectors as the columns of A .
2	Row reduce $A \rightarrow R$.
3	Count the pivots.
4	Pivots = number of vectors $\rightarrow$ independent . Fewer $\rightarrow$ dependent , and each special solution shows one dependency (which vector is a mix of which).

B. Basis and dimension of $C(A)$ and $N(A)$

Step	What to do
1	Row reduce $A \rightarrow R$ and mark the pivot columns.
2	Basis of $C(A)$: the pivot columns of A . $\dim C(A) = r$.
3	Basis of $N(A)$: the special solutions. $\dim N(A) = n - r$.
4	Check: $\dim C(A) + \dim N(A) = r + (n - r) = n$.

3. 2×2 examples

Budget and return only.

Example A: independent (a basis of $\mathbb{R}^2$)

Nifty $(1, 12)$ and bonds $(1, 7)$:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$$

- Row 2 $- 12 \times$ row 1: $(0, 7 - 12 = -5)$.
- Row 2 $\div (-5)$: $(0, 1)$. Row 1 $-$ row 2: $(1, 0)$.

$$R = \begin{bmatrix} \boxed{1} & 0 \\ 0 & \boxed{1} \end{bmatrix} \quad 2 \text{ pivots} = 2 \text{ vectors}$$

Therefore **independent**. Two independent vectors in $\mathbb{R}^2$ also span it, so Nifty and bonds are a **basis of $\mathbb{R}^2$** : every (budget, return) target is reachable in exactly one way. $\dim C(A) = 2$, $\dim N(A) = 0$.

Example B: dependent

Nifty $(1, 12)$ and its twin $(1, 12)$:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \xrightarrow{\text{row 2} - 12 \times \text{row 1}} R = \begin{bmatrix} \boxed{1} & 1 \\ 0 & 0 \end{bmatrix} \quad 1 \text{ pivot} < 2 \text{ vectors}$$

Therefore **dependent**. The special solution $(-1, 1)$ says $-1 \cdot \text{Nifty} + 1 \cdot \text{twin} = 0$: the twin is a copy. Both vectors lie on one line.

- Basis of $C(A)$: column 1 of A , $(1, 12)$. $\dim C(A) = 1$ (a line).
- Basis of $N(A)$: $(-1, 1)$. $\dim N(A) = 1$.
- Check: $1 + 1 = 2 = n$ ✓.

4. 3×3 examples

Example A: Nifty, bonds, midcap: a basis of $\mathbb{R}^3$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}$$

- $A \rightarrow U$: row 2 $- 12 \times$ row 1 $\rightarrow (0, -5, 3)$; row 3 $-$ row 1 $\rightarrow (0, -1, 0.2)$; row 3 $- 0.2 \times$ row 2 $\rightarrow (0, 0, 0.2 - 0.6) = (0, 0, -0.4)$.
- $U \rightarrow R$: row 3 $\div (-0.4) \rightarrow (0, 0, 1)$; row 2 $- 3 \times$ row 3 $\rightarrow (0, -5, 0)$, then $\div (-5) \rightarrow (0, 1, 0)$; row 1 $-$ row 2 $-$ row 3 $\rightarrow (1, 0, 0)$.

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad 3 \text{ pivots} = 3 \text{ vectors}$$

Therefore **independent**, and three independent vectors in $\mathbb{R}^3$ span it: a **basis of $\mathbb{R}^3$** . Every (budget, return, exposure) target has exactly one portfolio, e.g. $(10, 111, 7.4) \rightarrow (5, 3, 2)$ (Lesson 8). $\dim C(A) = 3$, $\dim N(A) = 0$.

Example B: Nifty, bonds, balanced: dependent

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix}$$

- $A \rightarrow U$: row 2 $- 12 \times$ row 1 $\rightarrow (0, -5, -2.5)$; row 3 $-$ row 1 $\rightarrow (0, -1, -0.5)$; row 3 $- 0.2 \times$ row 2 $\rightarrow (0, 0, 0)$.
- $U \rightarrow R$: row 2 $\div (-5) \rightarrow (0, 1, 0.5)$; row 1 $-$ row 2 $\rightarrow (1, 0, 0.5)$.

$$R = \begin{bmatrix} \boxed{1} & 0 & 0.5 \\ 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 \end{bmatrix} \quad 2 \text{ pivots} < 3 \text{ vectors}$$

Therefore **dependent**. The special solution $(-0.5, -0.5, 1)$ says balanced = 0.5 Nifty + 0.5 bonds.

- Basis of $C(A)$: pivot columns 1 and 2 of A : Nifty $(1, 12, 1)$ and bonds $(1, 7, 0)$.
 $\dim C(A) = 2$: the plane $R = 7B + 5E$.
- Basis of $N(A)$: $(-0.5, -0.5, 1)$. $\dim N(A) = 1$.
- Check: $2 + 1 = 3 = n$ ✓.

WHY NOT THE COLUMNS OF R?

R 's pivot columns are $(1, 0, 0)$ and $(0, 1, 0)$. They span the plane "third entry = 0". But $C(A)$ contains $(10, 105, 7)$ (₹7L Nifty + ₹3L bonds), whose third entry is 7. Different planes. Row operations changed the columns, so only A 's own pivot columns span $C(A)$.

▼ Check yourself (MIT problem 9.2): find a basis for the plane $x - 2y + 3z = 0$ in $\mathbb{R}^3$.

The plane is $N\left(\begin{bmatrix} 1 & -2 & 3 \end{bmatrix}\right)$. $R = \begin{bmatrix} 1 & -2 & 3 \end{bmatrix}$ already: pivot x , free y and z . Special solutions: $y = 1$ gives $x = 2$, so $(2, 1, 0)$; $z = 1$ gives $x = -3$, so $(-3, 0, 1)$. That's a basis; the plane has dimension 2.

5. Bigger example: five funds

All five funds, three rules. Five vectors in $\mathbb{R}^3$, so (concept 5) they **must** be dependent. How many genuinely different directions are there?

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 & 15 \\ 1 & 1 & 0 & 0.5 & 1.2 \end{bmatrix} \quad (\text{Nifty, ETF, bonds, balanced, midcap})$$

$A \rightarrow U$:

- Row 2 $- 12 \times$ row 1: $0, 0, 7 - 12 = -5, 9.5 - 12 = -2.5, 15 - 12 = 3 \rightarrow (0, 0, -5, -2.5, 3)$.
- Row 3 $- 1 \times$ row 1: $0, 0, -1, -0.5, 1.2 - 1 = 0.2 \rightarrow (0, 0, -1, -0.5, 0.2)$.
- Column 2 (ETF): nothing left to pivot on. Skip.
- Row 3 $- 0.2 \times$ row 2: $0, 0, 0, -0.5 + 0.5 = 0, 0.2 - 0.6 = -0.4 \rightarrow (0, 0, 0, 0, -0.4)$.
- Column 4 (balanced): nothing left to pivot on. Skip. Column 5 (midcap): pivot -0.4 .

$U \rightarrow R$:

- Row 3 $\div (-0.4)$: $(0, 0, 0, 0, 1)$.
- Row 2 $- 3 \times$ row 3: $(0, 0, -5, -2.5, 0)$; then $\div (-5)$: $(0, 0, 1, 0.5, 0)$.
- Row 1 $-$ row 3 $-$ row 2: $1, 1, 1 - 1 = 0, 1 - 0.5 = 0.5, 1 - 1 = 0 \rightarrow (1, 1, 0, 0.5, 0)$.

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 & 0 \\ 0 & 0 & \boxed{1} & 0.5 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

Pivot columns: 1, 3, 5 (Nifty, bonds, midcap). Free: 2, 4 (ETF, balanced). Rank = 3.

- **Independent?** No: 3 pivots < 5 vectors.
- **Basis of $C(A)$:** Nifty (1, 12, 1), bonds (1, 7, 0), midcap (1, 15, 1.2). Three independent vectors in $\mathbb{R}^3$, so $C(A) = \mathbb{R}^3$: $\dim C(A) = 3$.
- **Basis of $N(A)$:** special solutions $s_1 = (-1, 1, 0, 0, 0)$ (ETF = Nifty) and $s_2 = (-0.5, 0, -0.5, 1, 0)$ (balanced = half Nifty + half bonds). $\dim N(A) = 5 - 3 = 2$.
- Check: $3 + 2 = 5 = n$ ✓.

Check s_2 in A : budget $-0.5 + 0 - 0.5 + 1 + 0 = 0$ ✓; return $-6 + 0 - 3.5 + 9.5 + 0 = 0$ ✓; exposure $-0.5 + 0 + 0 + 0.5 + 0 = 0$ ✓.

5 funds, but only 3 genuinely different directions.

Drop the ETF and balanced and you lose **nothing**: Nifty, bonds, midcap still reach every target. That smallest complete menu is a basis.

Try it: pick a menu

☒ Nifty ☒ ETF ☒ Bonds ☒ Balanced ☒ Midcap

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 & 15 \\ 1 & 1 & 0 & 0.5 & 1.2 \end{bmatrix} \rightarrow R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 & 0 \\ 0 & 0 & \boxed{1} & 0.5 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

Rank $r = 3$ with 5 funds: **dependent**.

Basis of $C(A)$ (pivot columns of A): Nifty, Bonds, Midcap. $\dim C(A) = 3$: every target in $\mathbb{R}^3$ is reachable.

Basis of $N(A)$ (special solutions): $s_1 = (-1, 1, 0, 0, 0)$ (ETF is a mix of the funds before it); $s_2 = (-0.5, 0, -0.5, 1, 0)$ (Balanced is a mix of the funds before it). $\dim N(A) = 2$.

Check: $3 + 2 = 5$ funds.

6. Connection: rank, dimension and $n - r$

Every column of A is either a pivot column or a free column. So:

$$\underbrace{n}_{\text{funds}} = \underbrace{r}_{\dim C(A): \text{new directions}} + \underbrace{(n-r)}_{\dim N(A): \text{redundancies}}$$

Menu	n	$\dim C(A) = r$	$\dim N(A) = n - r$	Independent?
Nifty, bonds	2	2	0	yes
Nifty, twin	2	1	1	no
Nifty, bonds, midcap	3	3	0	yes (basis of $\mathbb{R}^3$)
Nifty, bonds, balanced	3	2	1	no
All five	5	3	2	no

How this ties the last lessons together:

Earlier idea	In this lesson's words
Column space $C(A)$ (Lesson 5)	the span of the columns; basis = pivot columns; dimension = rank
Null space $N(A)$ (Lesson 6)	basis = special solutions; dimension = $n - r$
Exactly one solution of $Ax = b$ (Lesson 8)	happens when the columns are a basis of the target space ($r = m = n$)
A do-nothing trade exists (Lesson 7)	the columns are dependent

READING IT AS A TRADER

$\dim C(A)$ is the number of **independent bets** the menu offers; adding funds beyond that adds no new exposure, only alternative routes. $\dim N(A)$ counts those alternative routes. A desk looking at 50 products driven by 3 risk factors has at most 3 independent bets, and at least 47 redundancies.

7. Final memory table

Concept	Meaning	Test / how to find it	Example
Independent	no vector is a mix of the others	rank = n (no free variables), $N(A) = \{0\}$	Nifty, bonds, midcap
Dependent	some non-zero mix = 0	rank < n ; special solutions show how	balanced = $\frac{1}{2}$ Nifty + $\frac{1}{2}$ bonds
Span	all combinations	columns of A span $C(A)$	Nifty, bonds span $R = 7B + 5E$
Basis	independent + spans	for $\mathbb{R}^n$: n pivots (invertible)	Nifty, bonds, midcap for $\mathbb{R}^3$
Dimension	vectors in any basis	count them	$\dim \mathbb{R}^3 = 3$
Basis of $C(A)$	reachable targets	pivot columns of A ; $\dim = r$	five funds: Nifty, bonds, midcap
Basis of $N(A)$	do-nothing trades	special solutions; $\dim = n - r$	five funds: s_1, s_2
Key count		$n = r + (n - r)$	$5 = 3 + 2$